

Varad Patil

varadap25@iitk.ac.in | LinkedIn | GitHub | Portfolio

Education

Year	Degree / Examination	Institute / Board	CPI / %
2027	M.Tech, AI for Sustainability	Indian Institute of Technology Kanpur	9.25
2025	B.Tech, Computer Engineering	Rajarambapu Institute of Technology	9.70

M Tech Thesis

Vision-Based PM2.5/PM10 Estimation from a City-Scale CCTV Network (ongoing) 2026

Advisor: Prof. Sachchida Nand Tripathi, Dean of Kotak School of Sustainability, IIT Kanpur

- Extracting physics-grounded features (RMS contrast, dark-channel prior, Koschmieder extinction, optical flow) alongside frozen **DINOv2 ViT** embeddings
- Fine-tuning per-camera **ResNet-18** regressors, benchmarked against a gradient-boosting ladder that attributes predictive contribution across optics, embeddings and meteorology
- Designing a leakage-controlled protocol with day-, week- and camera-blocked splits and blocked-bootstrap testing

Publications

A RAG-Based Legal Chatbot Using Llama2 and Advanced Information Retrieval 2024

First Author, International Conference on Trends in Computational and Cognitive Sciences (ICTCS 2024), Springer

- Engineered an end-to-end **retrieval-augmented generation** pipeline with **FAISS** vector indexing and **Llama2-7B** served via **LangChain**, designing corpus chunking, embedding strategies and prompt templates
- Built an evaluation framework measuring retrieval precision, latency and response quality across **500+** test queries

Professional Experience

Software Development Intern, Altair Engineering, Bangalore Jan–Jun 2025

- Built full-stack **MERN** client portals with a real-time **HPC job scheduling** and visualisation dashboard for tire manufacturers, surfacing live job status, queue position and simulation results
- Automated simulation and reporting workflows using **Python** and REST APIs, cutting manual operations by **40%**

Web Development Intern, Bubble Byte Ventures Pvt Ltd, Sangli Sep 2022–Aug 2023

- Delivered **10+** SEO-optimised client websites, cutting load times by **30%**; automated deployment pipelines in Python

Key Projects

CivicPulse: Multi-Agent Civic Issue Platform (Self Project) | Gemini 2.5-flash, Next.js 16, TypeScript, Firestore, Cloud Run

- Designed an orchestrated **multi-agent** pipeline: multimodal triage extracting category, severity and hazards, automated routing to municipal authorities with priority and SLA, and before/after repair verification by image comparison
- Applied a **minimal-AI architecture**, reserving inference for perception and judgement while keeping deduplication, geospatial hotspot prediction and routing deterministic
- Containerised with **Docker** and deployed on **Google Cloud Run** via Cloud Build, operating entirely within cloud free tiers

Resume Insight: LLM Resume Evaluation Engine (Self Project) | Next.js 16, React 19, Gemini API, Clerk, Supabase

- Engineered a **Gemini**-backed analysis service over PDF-extracted resume text, exposing separate endpoints for job-match scoring, ATS analytics and section-level rewriting with per-request token and error logging
- Built semantic resume to job-description matching with **ATS scoring**, keyword alignment and line-level improvement suggestions, deployed publicly with persistent analysis history
- Implemented tiered subscriptions on **Clerk** authentication and **Supabase** Postgres, enforcing per-feature monthly quotas, upload limits and automatic usage-period resets

CityLens: Urban Infrastructure Monitoring (Self Project) | YOLO, PyTorch, Ultralytics, SAHI, DDP

- Trained **YOLO** on the MBDD2025 UAV dataset (**14.5K** images, **57.6K** annotations) to **0.918 test mAP@0.5**, using iterative stratification and rare-class oversampling for a **10:1** imbalance, trained via **DDP** on dual T4 GPUs
- Built an illegal-billboard detector at **0.813 test mAP@0.5** through a five-variant ablation with a from-scratch AP@0.5 metric and validation/test gap tracking

Technical Skills

GenAI and LLM: RAG Systems, LangChain, FAISS, Llama2, Gemini, OpenAI API, Prompt Engineering, Multi-Agent Systems

Machine Learning and Deep Learning: PyTorch, TensorFlow, Transformers, CNNs, Object Detection, Scikit-learn, XGBoost

Languages and Backend: Python, TypeScript, FastAPI, Flask, Node.js, REST APIs

Data and Cloud: NumPy, Pandas, MongoDB, PostgreSQL, Firestore, GCP, Docker, Git

Certification: Generative AI with Large Language Models, DeepLearning.AI and AWS (2024)

Achievements

- Reliance Foundation Postgraduate Scholarship**, one of **100** awarded nationwide across all disciplines (2025)
- Gold Medalist**, graduated **Rank 1** in B.Tech Computer Engineering, Rajarambapu Institute of Technology (2025)
- Top 20 Finalist** of **15,000+** participants, Vibe2Ship National Hackathon; **Finalist**, CityLens AI Hackathon, IIT Kanpur (2026)
- Best Performance Award**, Bubble Byte Ventures Pvt Ltd, for consistent delivery across client projects (2023)

Positions of Responsibility

- Teaching Assistant**, Mathematics and Computation Using Python, IIT Kanpur, under Prof. Mahendra Kumar Verma (2026)
- Community Lead**, Google Developer Groups on Campus: led a technical community of **45+** members, organised **5+** workshops on web development and Generative AI, mentored **20+** students (2024–25)